

Can We Predict Every Pass? Building an xPass Model for Football

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XGBoost · StatsBomb open data · Event-only features · 360-enriched variant (2020/21 only)

Background

Just as Expected Goals (xG) estimates the probability of a shot resulting in a goal, the Expected Pass (xPass) framework predicts the likelihood of a pass being successfully completed based on the characteristics of the pass and the surrounding game context. In essence, xPass serves as a measure of pass difficulty by estimating the probability of completion given the state of play. xPass models have a wide range of applications, including separating decision quality from execution quality, evaluating passing ability, player scouting, and tactical analysis. Existing literature primarily focuses on features such as player orientation, opponent positioning, ball trajectory, and spatial context. However, one potentially influential factor that has received little attention is a player's stronger foot. In this work, we investigate does a player's weak foot hurt pass completion, and how well-calibrated is the model across different pass types?

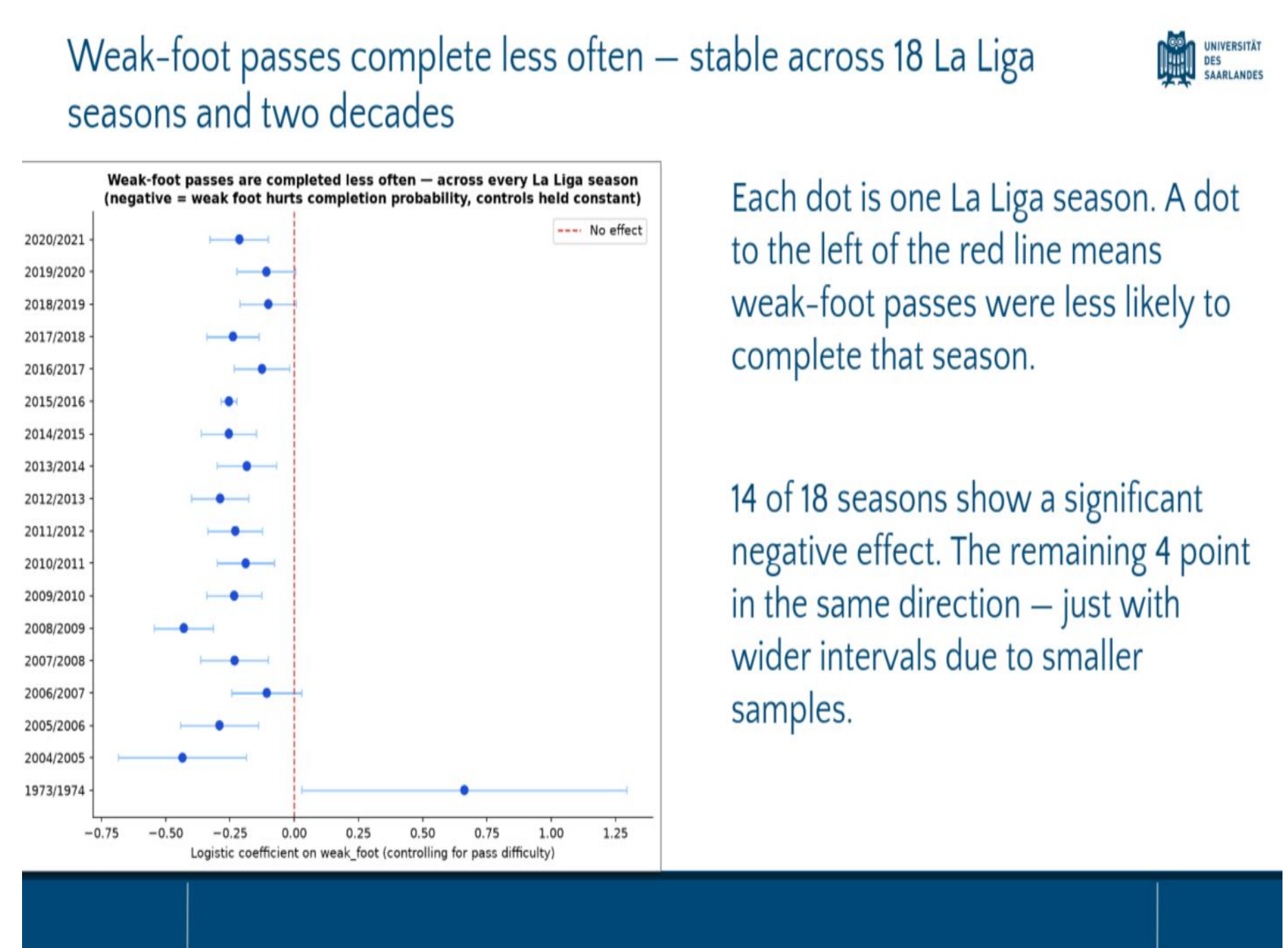


Fig. 1. Per-season logistic regression coefficient on weak-foot flag, controlling for pass length, pressure, cross/switch/through-ball flags and distance to goal (95% CI shown). Negative values indicate weak-foot passes are less likely to complete after controlling for difficulty. Significant in 14 of 18 seasons; 1973/74 excluded from headline claim (n=469, incomplete body-part tagging).

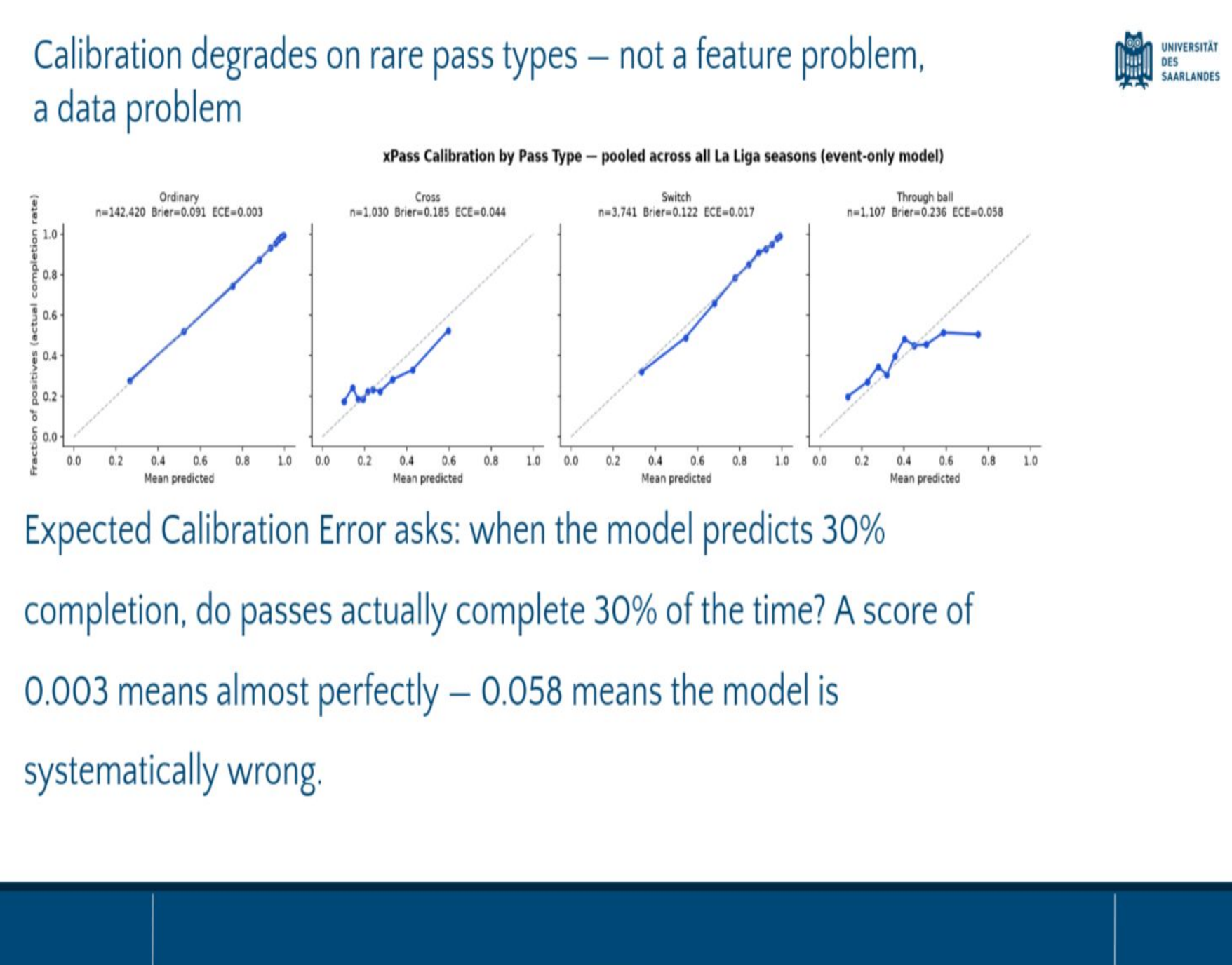


Fig. 2. Expected Calibration Error (ECE) by pass type, pooled across 18 La Liga seasons. Lower ECE = better calibration. The model is near-perfectly calibrated on ordinary passes (n=142,420, ECE=0.003) but degrades sharply on rare types — through balls (ECE=0.058) are 19× worse calibrated than ordinary passes. Degradation scales directly with sample scarcity, not pass type per se.

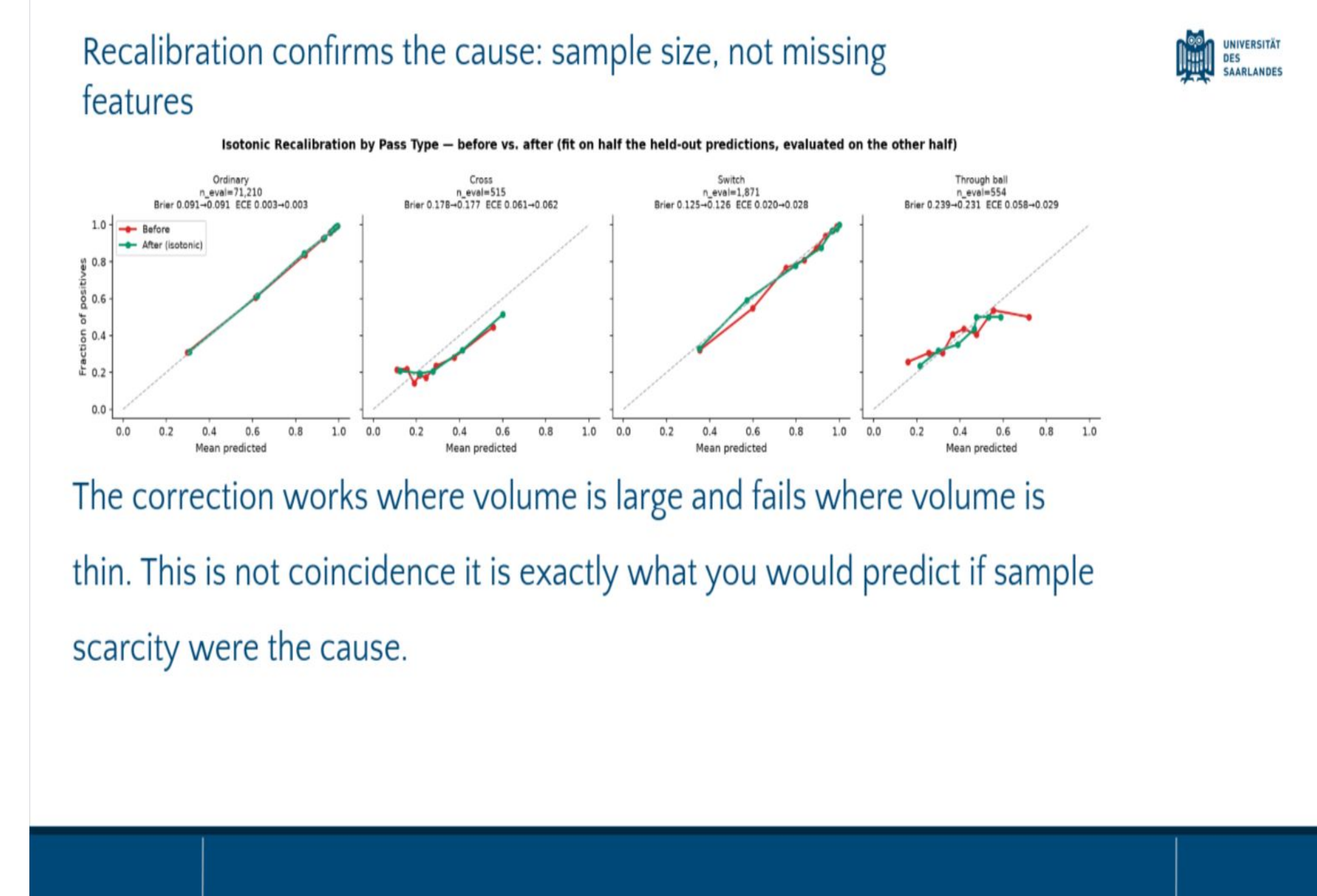


Fig. 3. Isotonic regression recalibration applied to held-out predictions per pass type. Correction fitted on one half of held-out data, evaluated on the other. The correction succeeds where volume is sufficient and fails where data is thin confirming that sample scarcity, not missing features, is the cause of poor calibration on rare pass types.

Methods

Data: StatsBomb open-data events, 18 La Liga seasons (1973/74–2020/21), 400,000+ passes. Preferred foot scraped and inferred

Features: pass length, angle, distance/angle to goal, forward progress, pitch zone, pass height, under pressure, cross/ switch/ through-ball flags, weak-foot flag, weak-foot × cross interaction.

Model: XGBoost per season, match-level 80/20 split.

Isotonic regression: Isotonic regression recalibration fits a monotone step function mapping model predictions to observed outcomes. Flexible on large samples, prone to overfitting on small ones. We use this property as a diagnostic: if the correction succeeds where n is large and fails where n is thin, that pattern directly confirms sample scarcity as the cause of poor calibration.

Results

Weak-foot effect (Finding 1):

Typical effect: -20–25% lower odds of completion controlling for pass difficulty.

Calibration by pass type (Finding 2):

Ordinary passes: ECE 0.003, near-perfect. Through balls: ECE 0.058, 19× worse. Model is overconfident on every rare type. Pattern scales directly with sample scarcity. Overall AUC 0.85–0.90 — model is strong globally, not locally.

Ablation (Finding 3):

Footedness features give marginal improvement; more footedness interactions overfit. Missing features are not the cause.

Recalibration (Finding 4)

Isotonic correction halves ECE on through balls (n=554) but worsens it on crosses (n=515) and switches (n=1,871). Success and failure scale exactly with n. Direct confirmation: sample size is the root cause, not feature engineering.

Discussion

- Player footedness is a meaningful predictor of pass success, with weak-foot passes showing 20–25% lower completion odds across most La Liga seasons, even after controlling for pass difficulty.
- Footedness improves xPass predictions but does not fully explain calibration errors.
- Adding player-specific technical attributes makes xPass models more personalized and useful for player evaluation, scouting, and tactical analysis.
- Poor calibration on rare pass types is driven primarily by data scarcity, not missing player features.
- Future work includes richer player features and improved calibration techniques for low-frequency events.